Sociotechnical Security

CS 8803/4803

Professor Veronica Rivera

Lecture 14 – Online Hate and Harassment



Content warning: mentions and examples of hateful online content and harassment against vulnerable groups.

Last time

- Approaches to studying scams in digital safety
- Role of AI in enabling romance-baiting scams
- Discussion on ways of educating/training older adults and other groups to identify scams

Today

- Very very high-level overview of research in online hate and harassment
- Additional taxonomy
- Areas of work
- Solution spaces
  - How some fall short depending on context
- Muiz leading discussion

Definition

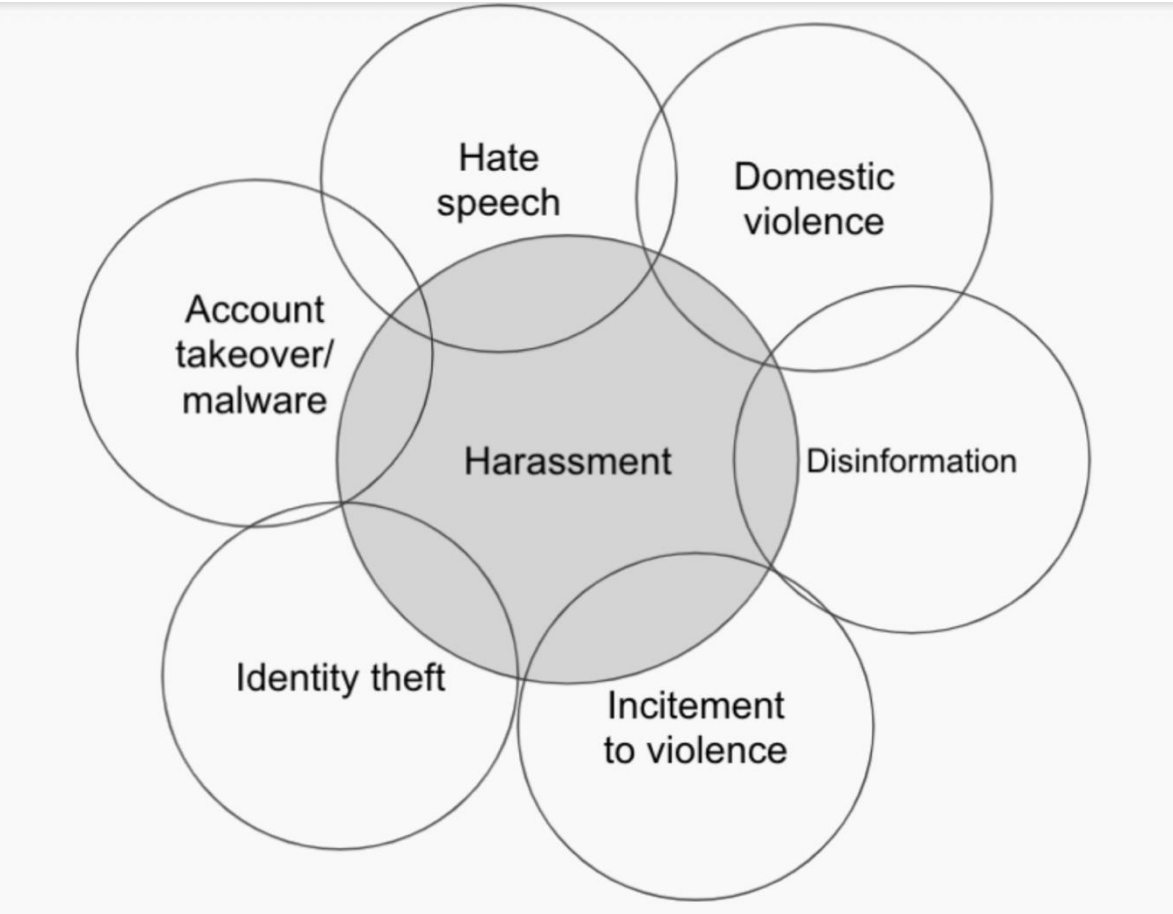


Figure credit: T&S textbook https://tsbook.org/ch6-harassment/#online-harassment

Online hate and harassment: the use of online technologies to intimidate, demean, silence, or endanger people.

Hate refers to attacks motivated by someone's group identity.

Harassment refers to targeting a specific person, whether through content (slurs, threats), through coordination (dogpiling, brigading), or through actions that cause harm offline (doxxing, swatting).

The two often overlap: a person is targeted because of who they are.

Situating hate and harassment

Usable security and privacy

Digital safety

Overlap



Authentication

Scams

Hate and harassment

Problem flavor: We can make computer systems that are provably secure against an adversary, but human behavior reduces these guarantees. How do we make humans behave more securely?

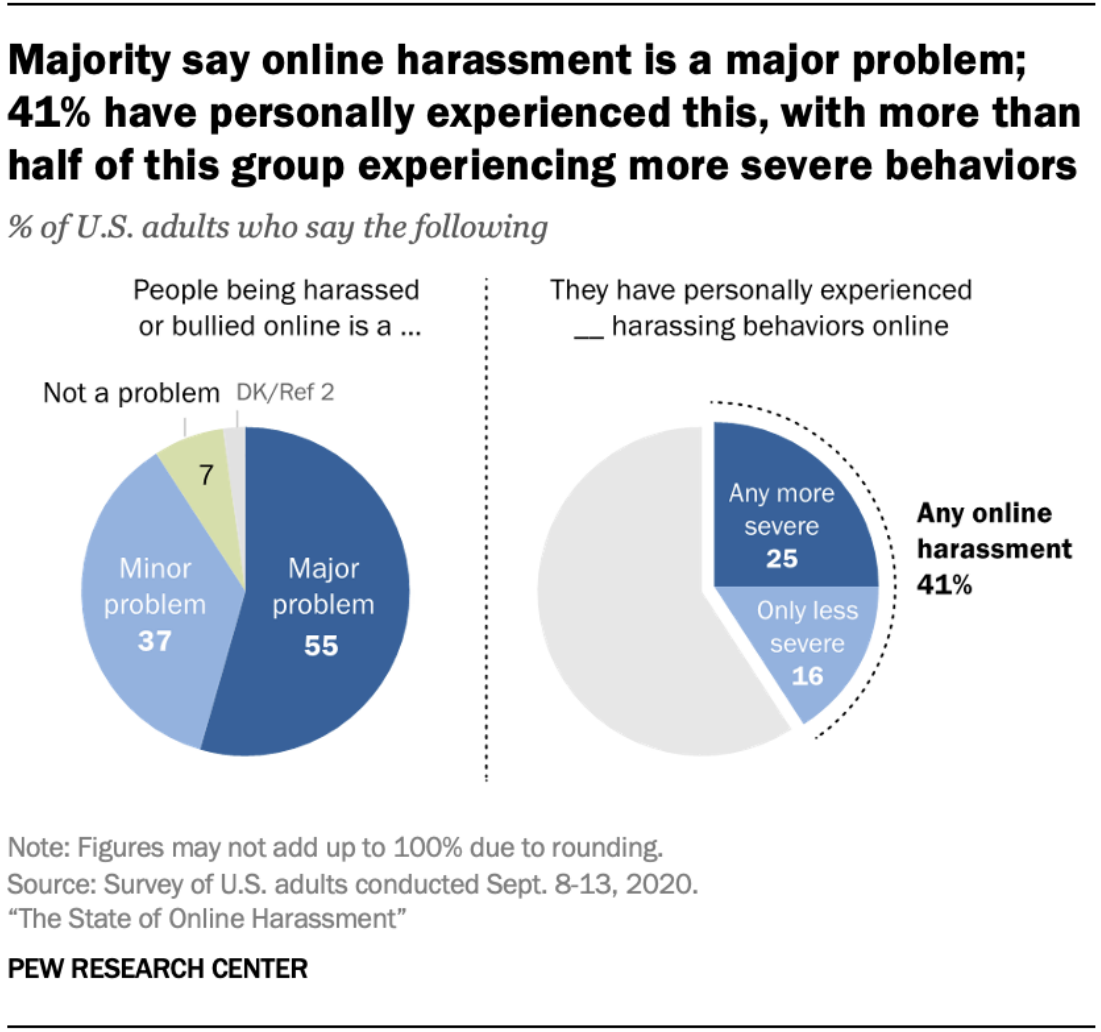
Problem flavor: How are computer systems used to cause harm and what how can we describe these problems?

Example: Gamergate (2014-2015)

A misogynistic online harassment campaign targeting feminism and diversity in video game culture

- Main targets included female media critics and game developers, beginning with Zoë Quinn
- A false blog allegation in August 2014 sparked a campaign that spread across online forums
- One of the first examples illustrating how online harm has severe offline impacts
- Cultural conflict centered progressiveness in themes underlying games

Hate & harassment is highly prevalent



- Pew Research 2021 study:
  - 55% believe online harassment is a problem
  - 41% of surveyed US Americans have experienced online harassment
  - Over half have experienced **severe** harassment
- UNESCO 2023 study:
  - 72,409 videos violated YouTube hate & abusive policy in Jan-June 2023
  - 9.6 Million pieces of content violated Meta's hate speech policy in Jul-Sep 2023

Taxonomies of online hate and harassment

Taxonomies are great for helping us understand complex topics that are studied in different research communities

One example: Kurt Thomas et al., SoK: Hate, Harassment, and the Changing Landscape of Online Abuse

- Broken down by intended audience, communication channel used in the attack, resulting harms

Another: Trust & Safety Online Textbook

- Broken down by where the harassment happens and how many people take part

Three categories of online harassment

Category	Brief definition	Example from the textbook
Public	Harassment in public spaces, which a single person can carry out.	Doxxing: A website exposed activist Ruslan Leviev's personal information.
Communal	Coordinated harassment by a group in public spaces.	Brigading: Restaurants receive waves of negative reviews after a political candidate visits.
Private	Harassment directed at an individual through private, one-to-one interactions.	Individual threats: Death threats sent through messaging apps, where there may be less filtering of abuse.

Source: Trust and Safety: An Open Source Textbook, Chapter 6 (examples paraphrased)

Areas of prior work on hate & harassment

Prior work identifies and measures the shape of the problem:

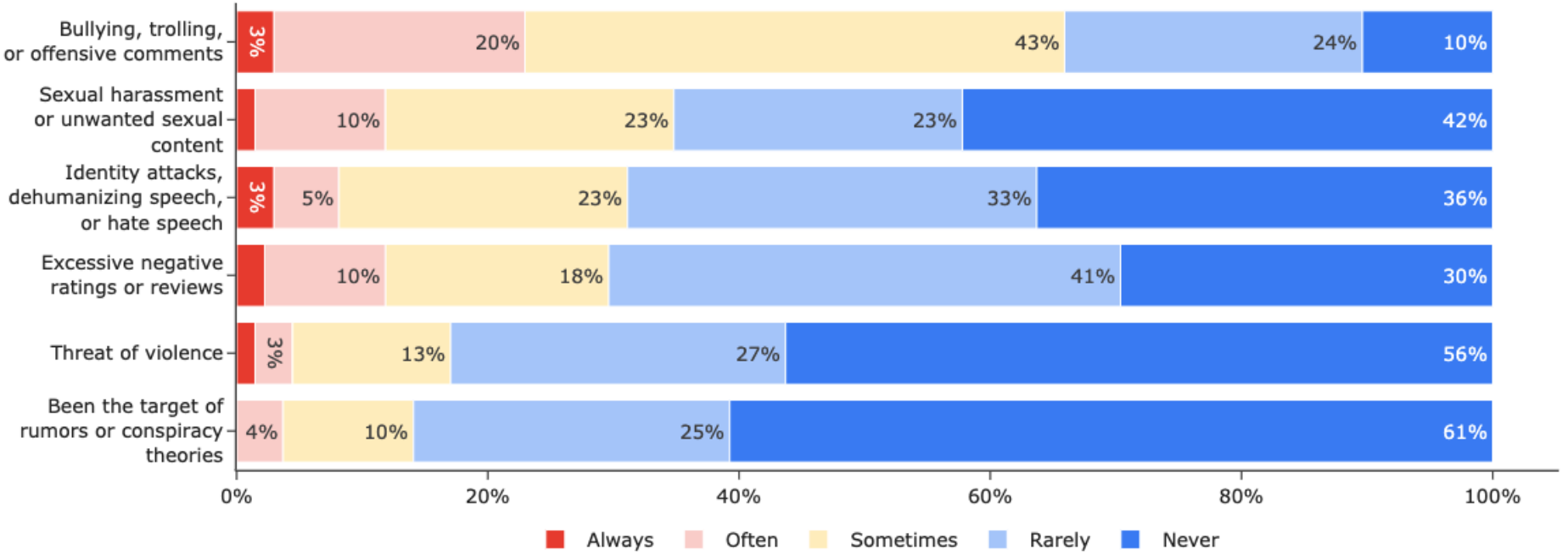
- Prevalence of hate and harassment
- How attacks spread
  - Example: Measurement of attacks on hate raids on twitch (Catherine Han et al., CSCW 2024)
- Who is most vulnerable
- (More recent) Generative AI as an attack tool
  - Automated hate campaigns (Xinyue Shen et al., USENIX Security 2025)
  - Harassment by AI agents
  - Image-based sexual abuse

Vulnerable groups

- Content creators
  - Next two slides: Kurt Thomas et al., “It’s common and part of being a content creator”: Understanding How Creators Experience and Cope with Hate and Harassment Online, CHI 2022
- Journalists
- Researchers and academics
- Political campaigns
- Youth
- Streamers and gamers

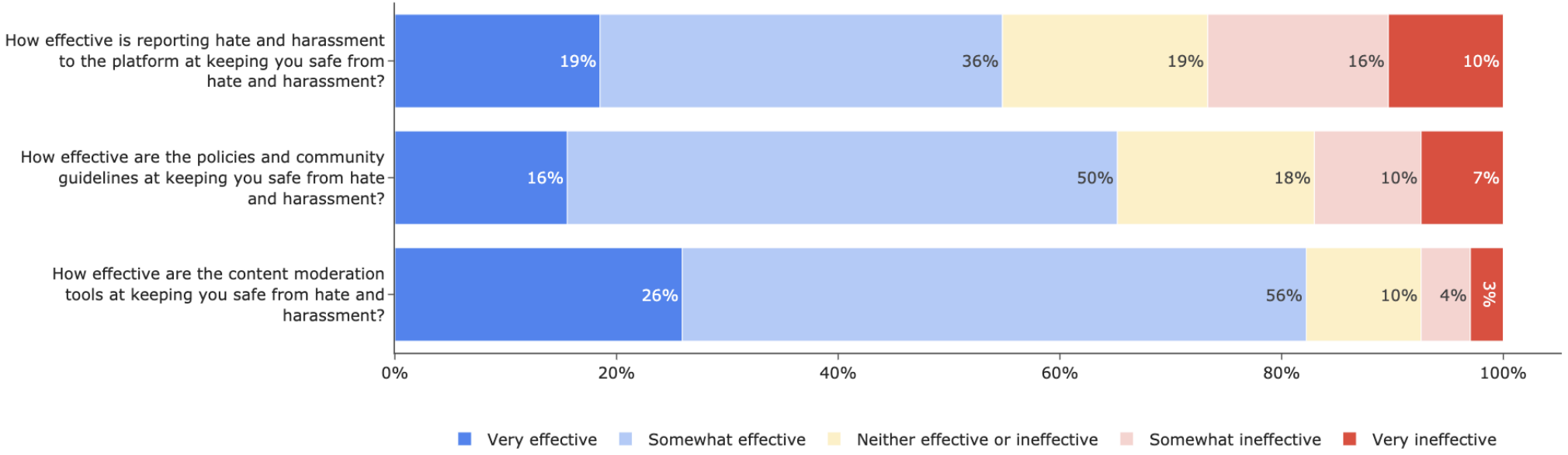
Content creators' experiences with h&h

- 95% of creators reported experienced some form of hate and harassment, higher than the prevalence among US adults
- Bullying, trolling, and offensive comments, occurred more frequently than other attacks
- Multiple platforms involved
- Attackers: audience (42%), creators (23%), repeat offenders (20%), and groups (17%).
- Impact: Self-censorship and avoidance



Content creators' attitudes towards solutions

- Reporting considered least effective, moderation considered most effective
- Confusion around reporting workflows reduces their use among this group
- Risk of moderation going too far
- Top creator recommendations:
  - Preventing anonymous or low-quality accounts from posting comments (56%)
  - Automatic warnings to commenters (54%)
  - Improved platform reporting tools (50%)



Solution spaces

- Moderation
- Post-harm support
  - In HCI especially: Restorative approaches (e.g., what do victims want beyond having content removed)?
- Design
  - Heartmob: interface for reporting hate and harassment to bystanders
    - Blackwell et al., CSCW 2017
  - Squadbox: tool for family and friends to step in and moderate toxic email content on behalf of a target
    - Mahar et al., CHI 2018

Different types of moderation

- Platform moderators
  - Paid moderators enforcing policies centrally; harm on moderators' mental health
- Algorithmic moderation
  - Toxicity classifiers for known harmful content
- Community moderators
  - Moderators of subreddits, Discord servers, or Twitch channels. They understand local norms, but the labor is unpaid and they become targets themselves.

One size fits all moderation doesn't work

- Some groups must engage with potentially harmful content to do their jobs
  - Journalists often avoid blocking or muting harassers because it can signal to attackers that the harassment worked and conflicts with their duty to keep their work accessible to the public.
  - Content creators may still want to engage with discussions expressing differences of opinion

Han et al., PressProtect: Helping Journalists Navigate Social Media in the Face of Online Harassment, CSCW 2024.

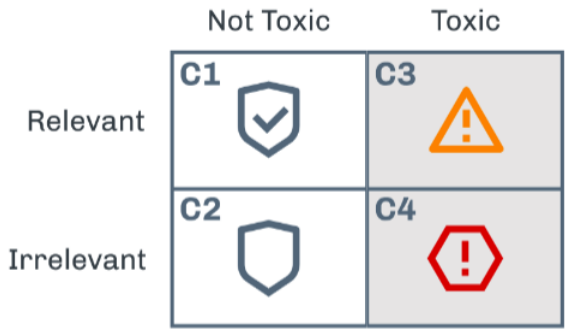
1. What are journalists' specific online safety needs?
2. What decisions do they make when interacting with hateful content?
3. What kind of tool can enable journalists to use social media in the face of harassment?

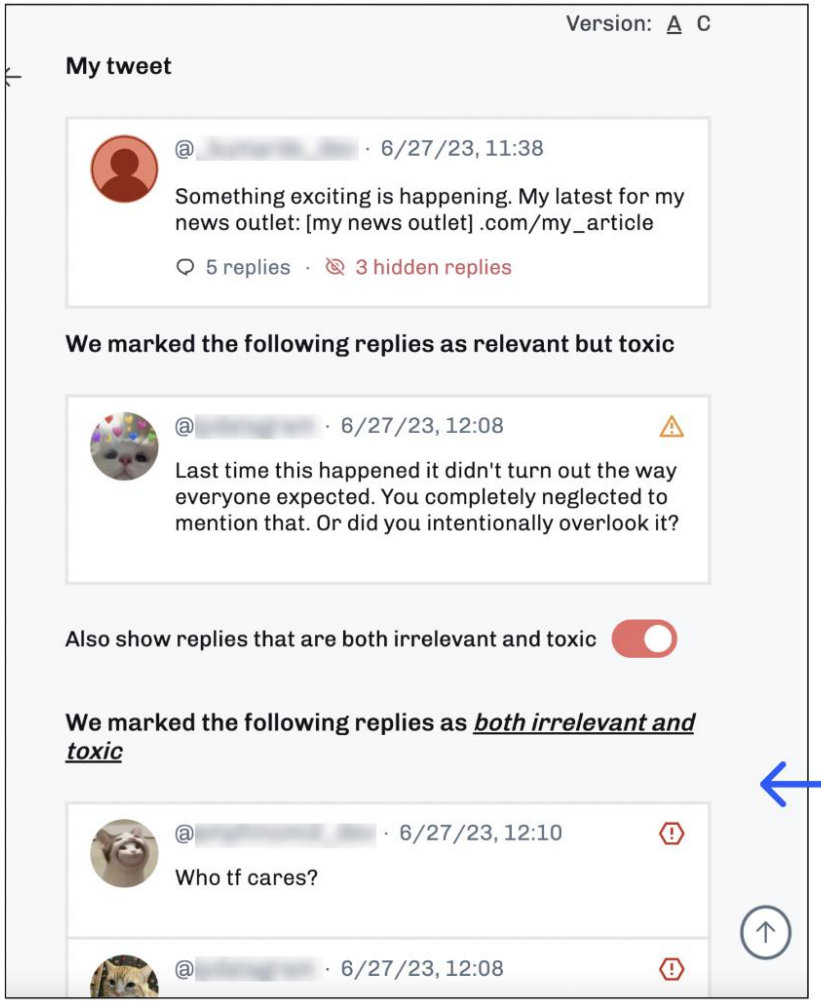
PressProtect

Classifies each reply by how relevant it is to the journalist's post or story, and how toxic it is

- Not toxic/relevant:
  - "Thanks for your article, but you used the wrong term for this illness."
- Not toxic/relevant:
  - "Geez"
- Toxic/relevant:
  - "They did terrible things, yet you wrote a glowing story. What a disgrace."
- Toxic/irrelevant:
  - "No one gives a ****, this is 2023."

- New need emerged: flagging physical threats
- Future work: classify harassment based on identity, beat, newsroom support, and past experiences





4. Replies page (harmful): C3 and C4

Open problems

- What specific design features in online platforms contribute to online hate and harassment?
- How to measure success?
- Will generative AI create new threat models for online hate and harassment?
- What role could/should generative AI play in moderation?
  - E.g., On what types of content is it likely to fail where human moderator should step in?

Reminders:

- No class on Monday, but do still turn in your midpoint progress reports
- Sign up for Volkan's talk next Thursday if you will attend
- IMPORTANT: If you will attend and do not have Coda access, you need to let me know by Friday via email